

Lesson: Lists in Python



Introduction

In Python, a **list** is used to store multiple values in a single variable.

Instead of creating separate variables like this:

You can store all values in one place:

Lists are one of the most important data structures in Python. They are used to:

- store collections of data
- manage groups of related items
- loop through data
- build real-world applications like apps, games, and databases

```
student1 = Alice  
student2 = Bob  
student3 = Charlie
```

```
students = [Alice,  
Bob, Charlie]
```

What Is a List?

A list is:

- **ordered** (items have positions)
- **mutable** (you can change it)
- written using **square brackets** `[]`



```
numbers = [1, 2, 3, 4, 5]
```

Accessing List Elements

Each item in a list has an **index** (position).
Indexes start at **0**.



```
fruits = [apple, banana, orange]
print(fruits[0])    # apple
print(fruits[1])    # banana
print(fruits[2])    # orange
```

Modifying List Items

You can change values in a list:

```
fruits = [apple, banana, orange]  
fruits[1] = grape  
print(fruits)
```

Output:

```
['apple', 'grape', 'orange']
```


Adding Items to a List

1. Append (add to end)



```
fruits = [apple, banana]  
fruits.append(orange)  
print(fruits)
```

2. Insert (add at a specific position)



```
fruits.insert(1, grape)  
print(fruits)
```


Removing Items from a List

1. Remove by value



```
fruits = [apple, banana, orange]  
fruits.remove(banana)  
print(fruits)
```

2. Remove by index



```
fruits.pop(1)  
print(fruits)
```

Looping Through a List

Lists are often used with loops.

Example: Print all items



```
fruits = [apple, banana, orange]
for fruit in fruits:
    print(fruit)
```

List Length

Use `len()` to count items:



```
numbers = [10, 20, 30, 40]  
print(len(numbers)) # 4
```

Checking if an Item Exists



```
fruits = [apple, banana, orange]
if banana in fruits:
    print(Banana is in the list)
```

Lists with Different Data Types

Python lists can store mixed data types:



```
data = [Alice, 25, True]
print(data)
```


Real-World Use Cases

1. Storing Student Names



```
students = [Alice, Bob, Charlie]
for student in students:
    print(fWelcome, {student}!)
```

2. Tracking Scores



```
scores = [85, 90, 78, 92]
total = 0
for score in scores:
    total += score
average = total / len(scores)
print(Average score:, average)
```


3. To-Do List Application



```
tasks = []
tasks.append(Study Python)
tasks.append(Go to the gym)
tasks.append(Read a book)
for task in tasks:
    print(task)
```

4. Shopping Cart



```
cart = [milk, bread, eggs]
if milk in cart:
    print(Milk is in your cart)
```

Combining Lists with If Statements

Lists work very well with decision-making.



```
numbers = [1, 2, 3, 4, 5]
for num in numbers:
    if num % 2 == 0:
        print(num, is even)
    else:
        print(num, is odd)
```

Common Beginner Mistakes

Index out of range

```
fruits = [apple, banana]  
print(fruits[2]) # ERROR
```

Fix:

Check length first:

```
if len(fruits) > 2:  
    print(fruits[2])
```

Forgetting parentheses on methods

```
fruits.append    # WRONG
```

 **Correct:**

```
fruits.append(orange)
```

Full Program Example



```
students = []
# Add students
students.append(input(Enter student name: ))
students.append(input(Enter another student name: ))
print(\nStudent List:)
for student in students:
    print(student)
# Check if a student is in the list
search = input(\nEnter a name to search: )
if search in students:
    print(search, is in the list.)
else:
    print(search, is not in the list.)
```


Why Lists Matter

Lists are essential because they allow you to:

- store and organize multiple values
- process groups of data efficiently
- build dynamic programs
- support features like loops, conditions, and user input

Without lists, programs would be limited to handling only one value at a time.

Summary

In this lesson, you learned:

- A list stores multiple values in one variable
- Lists are ordered and mutable
- Use indexes to access items
- Add items with `append()` and `insert()`
- Remove items with `remove()` and `pop()`
- Use loops to process list data
- Use `len()` to count items
- Use `in` to check for values

Lists are one of the most important tools in Python and are used in almost every real-world program.